

Ethanol Blending in Gasoline - Argentina

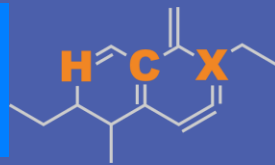
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Argentina

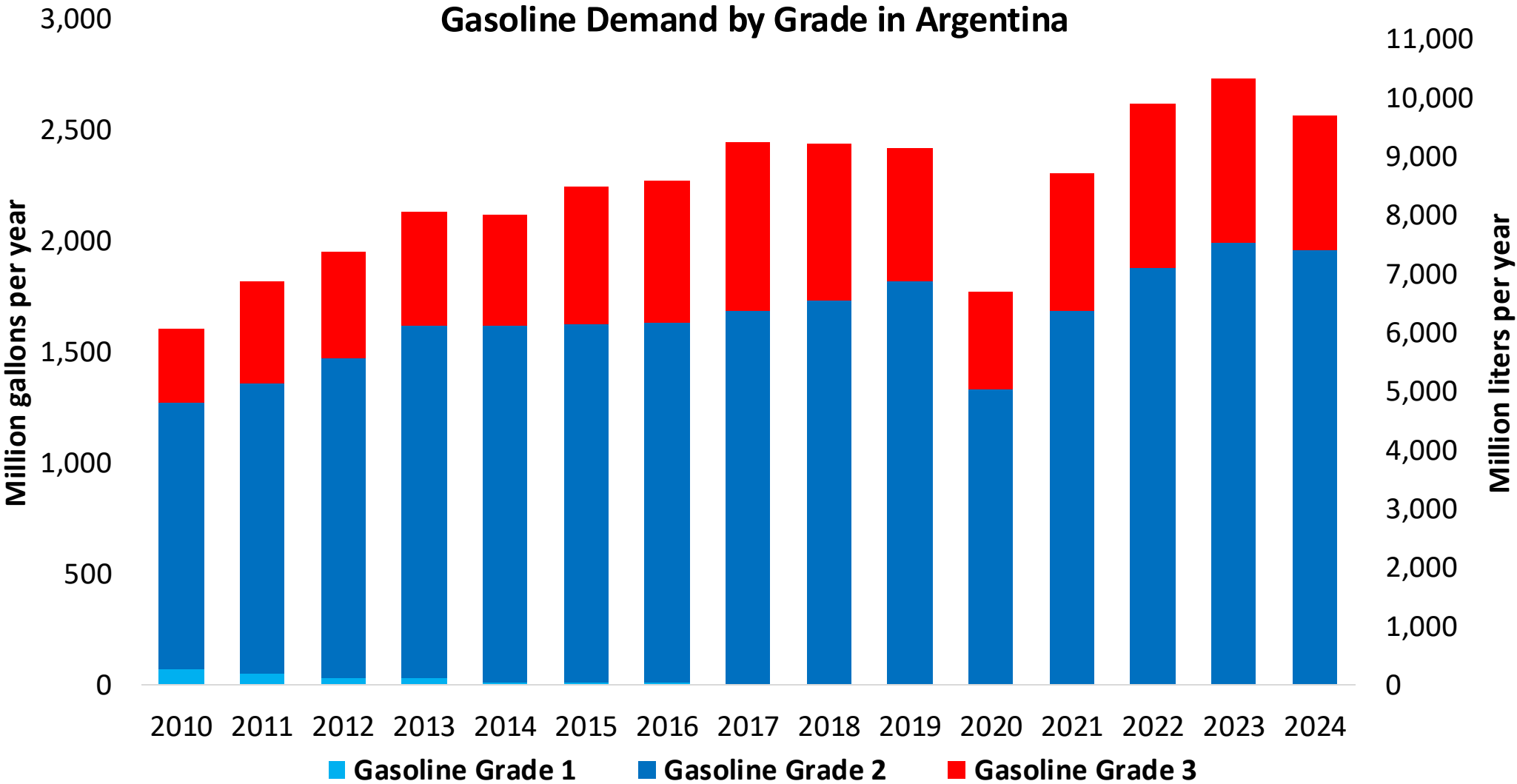


Argentina's total gasoline demand was 2,566 million gallons per year in 2024 (9,712 million liters per year), with a market share of approximately 76% as Nafta Súper / Grado 2 (RON 93 min – AKI 87) and 24% as Nafta Ultra / Grado 3 (RON 97 min – AKI 92). Octane delivered to market is usually higher than the minimum required. The demand for gasoline is supplied by local production, requiring imports of less than 75 million liters per year.

Argentina's policy is self-sufficiency and production and supply of ethanol for gasoline blend. Ethanol concentration varies according to local production. Currently, the concentration limit is 12% v/v. (Law 27.640) and a future E15 requirement is under discussion. Ethanol production is equal to consumption with some exports and imports from other countries.

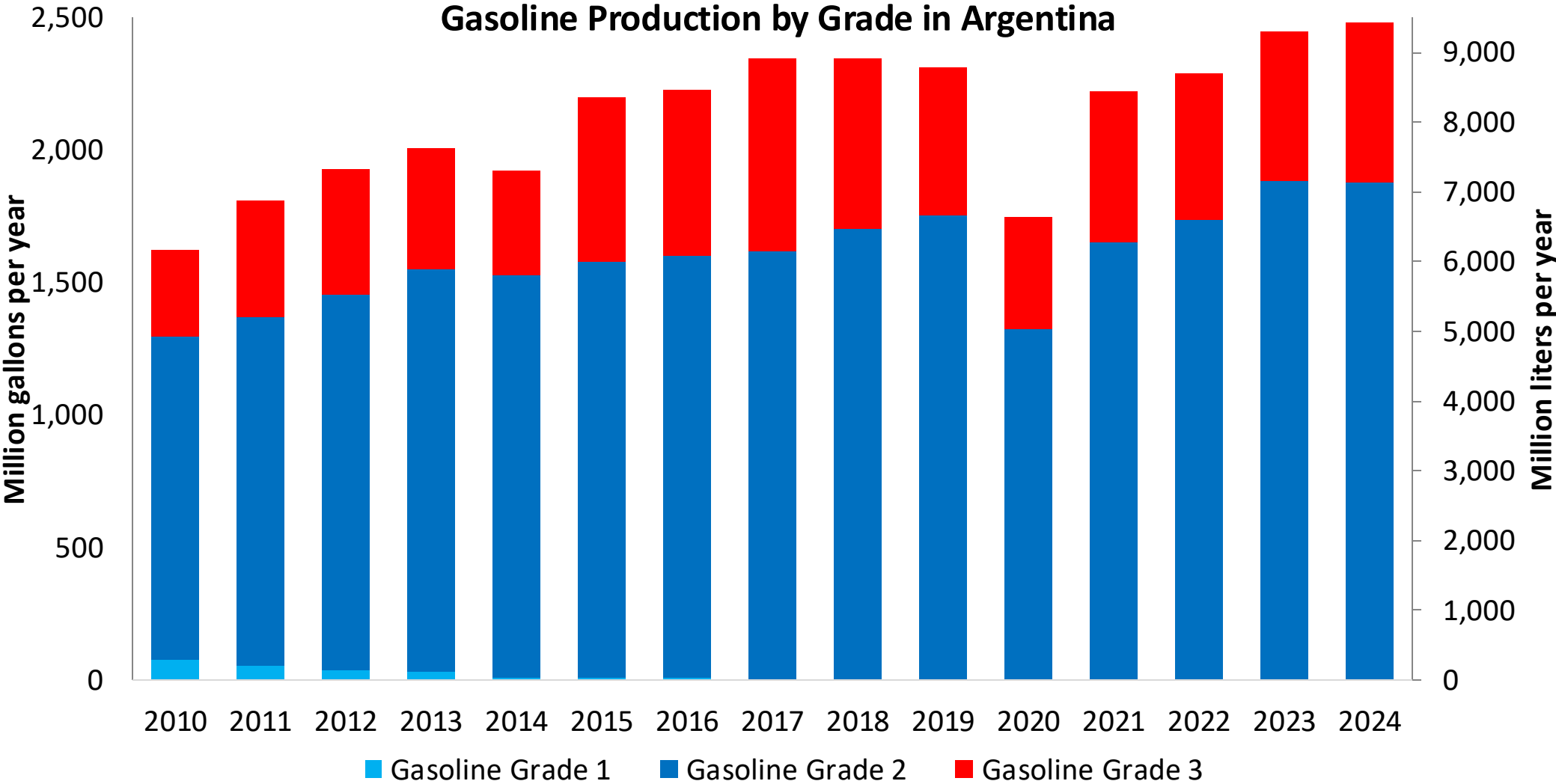
Source: Ministerio de Economía, 2025

Gasoline Demand in Argentina



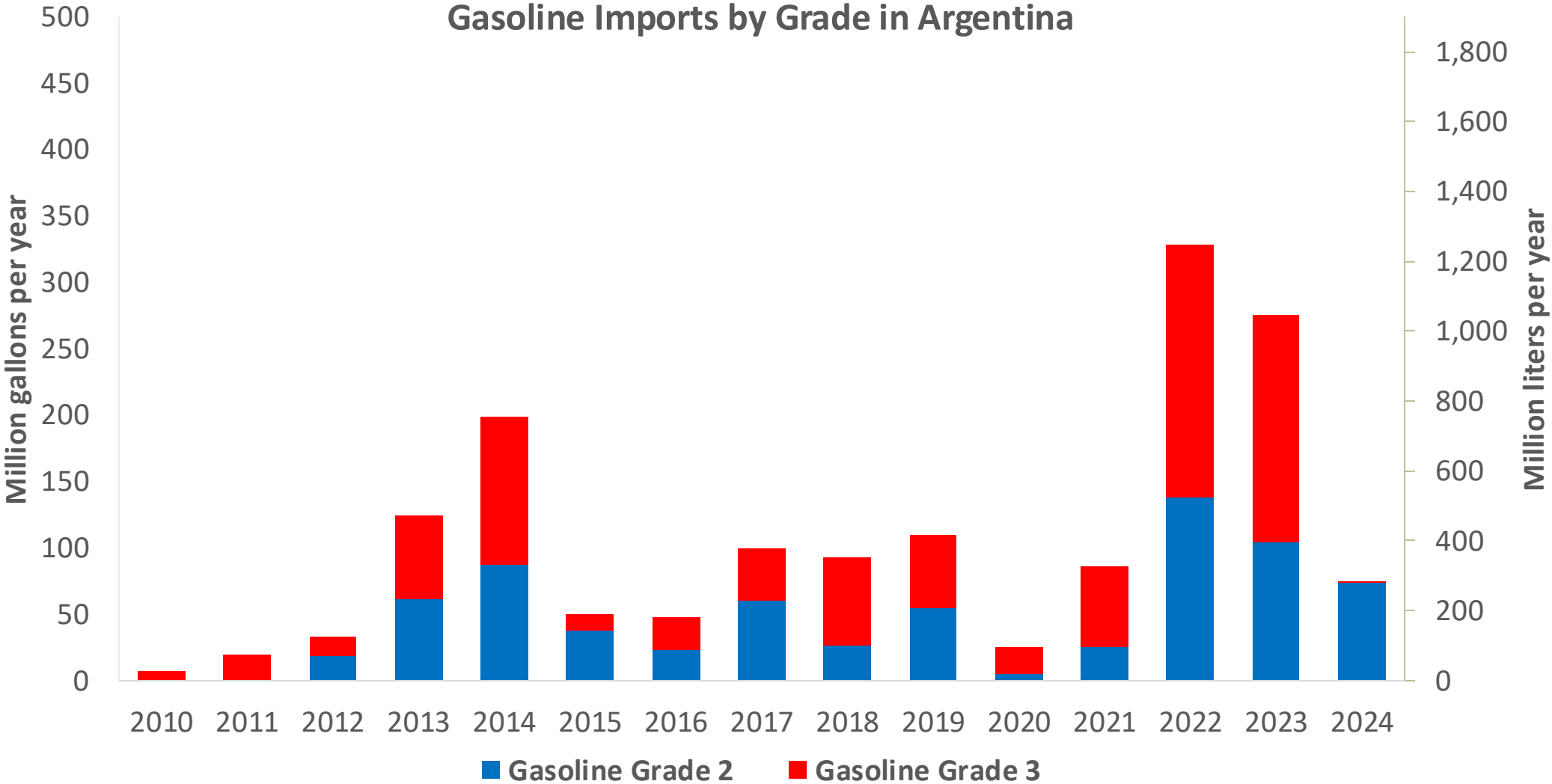
Source: Ministerio de Economía, 2025

Gasoline Production in Argentina



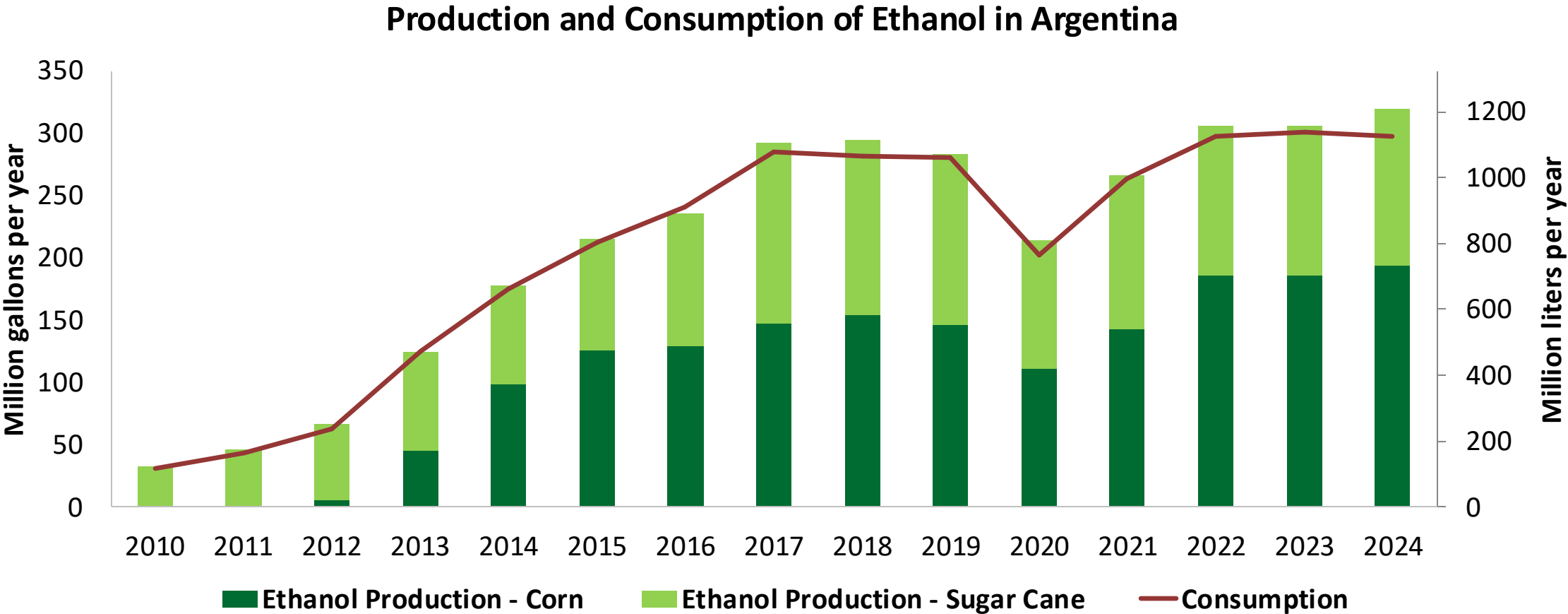
Source: Ministerio de Economía, 2025

Gasoline Imports in Argentina



Source: Ministerio de Economía, 2025

Ethanol Balance in Argentina



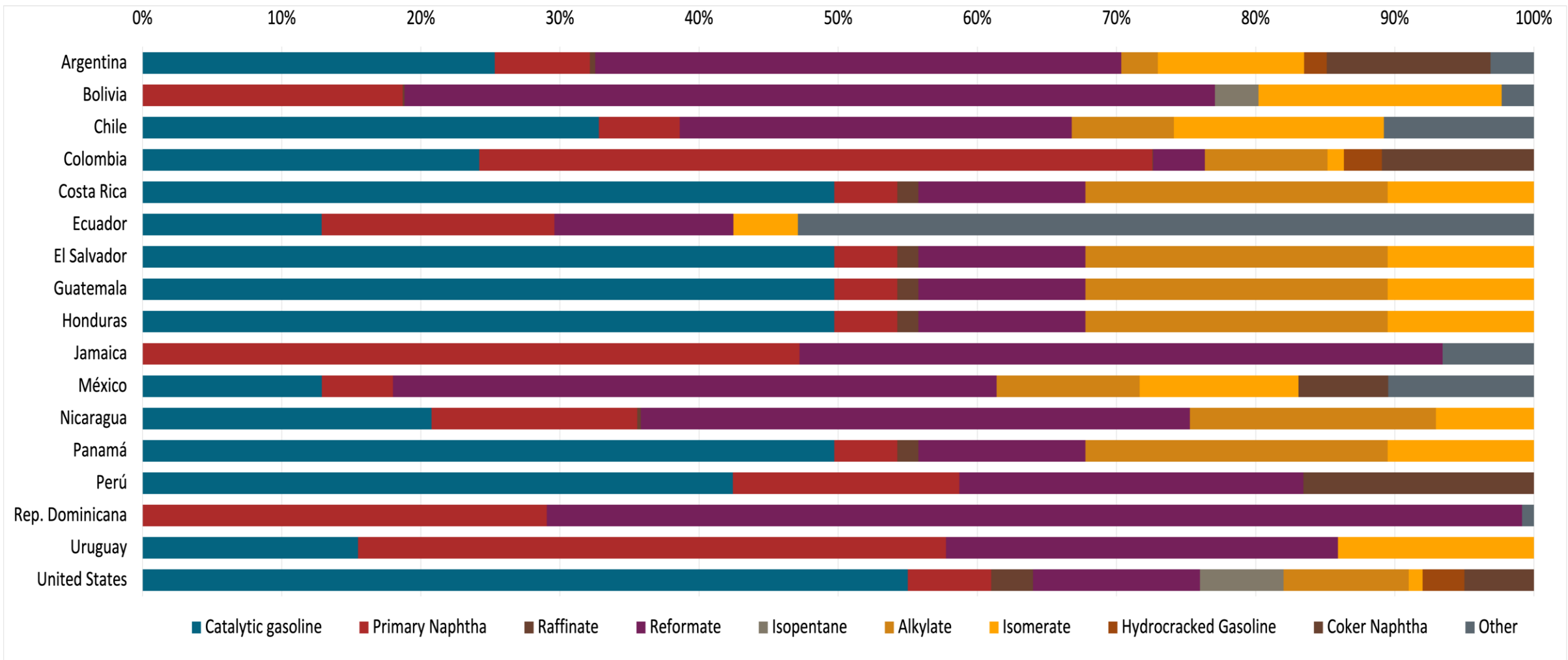
Source: Ministerio de Economía, 2025

Gasoline Quality in Argentina

Name	Resolution 576/2019 and Resolution 492/2023						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019						2017			
Applicability	North Zone		Central Zone		South Zone		All countries			
Selected Grade	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 93	> 97	> 93	> 97	> 93	> 97	> 95	> 95	> 98	> 98
MON	> 83	> 85	> 83	> 85	> 83	> 85	> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 12 - 14 %v/v	< 12 - 14 %v/v	< 12 - 14 %v/v	< 12 - 14 %v/v	< 12 - 14 %v/v	< 12 - 14 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 35 - 70 kPa	< 35 - 70 kPa	< 35 - 70 kPa	< 35 - 70 kPa	< 45 - 80 kPa	< 45 - 80 kPa	< 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	< 45 - 80 kPa	< 45 - 80 kPa	< 55 - 90 kPa	< 55 - 90 kPa	< 55 - 90 kPa	< 55 - 90 kPa				
RVP 37.8°C (Transition)			< 45 - 80 kPa	< 45 - 80 kPa						
MTBE	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	-	-	-	-
Ethers 5 or more C Atoms	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



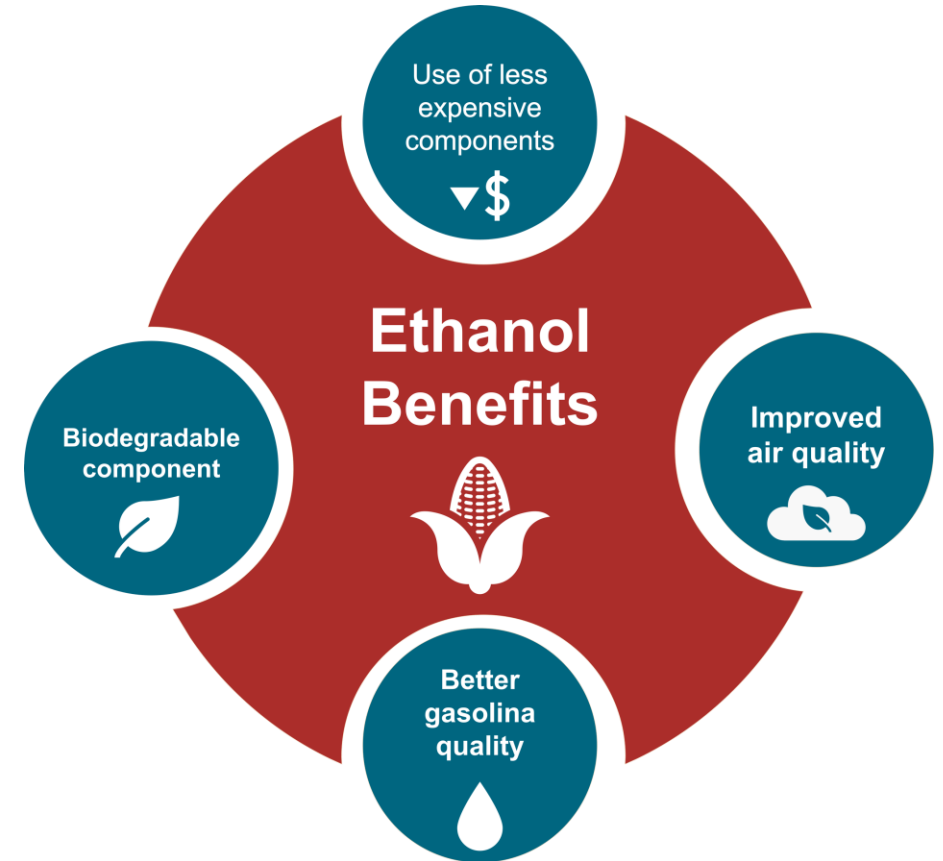
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

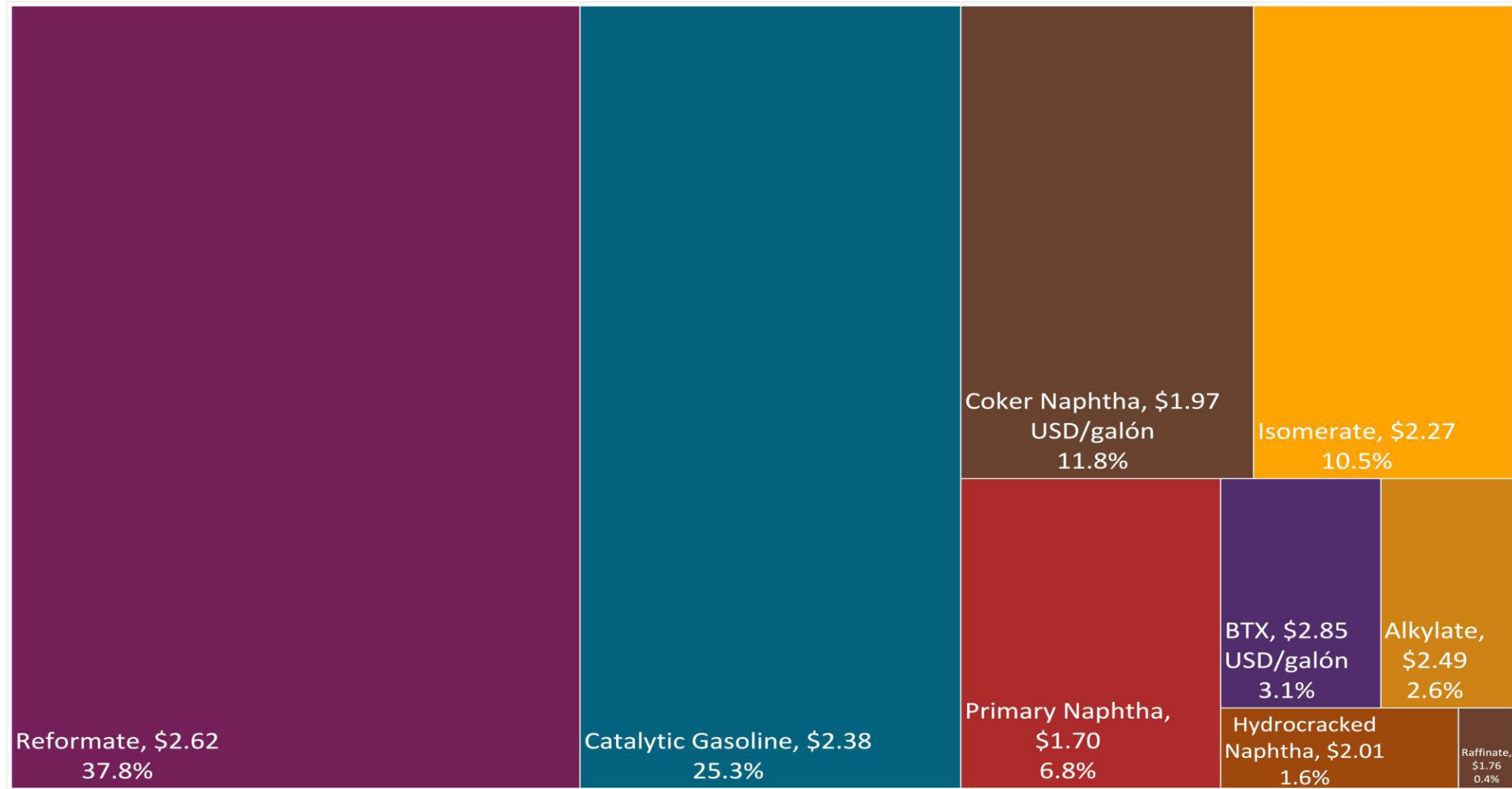
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

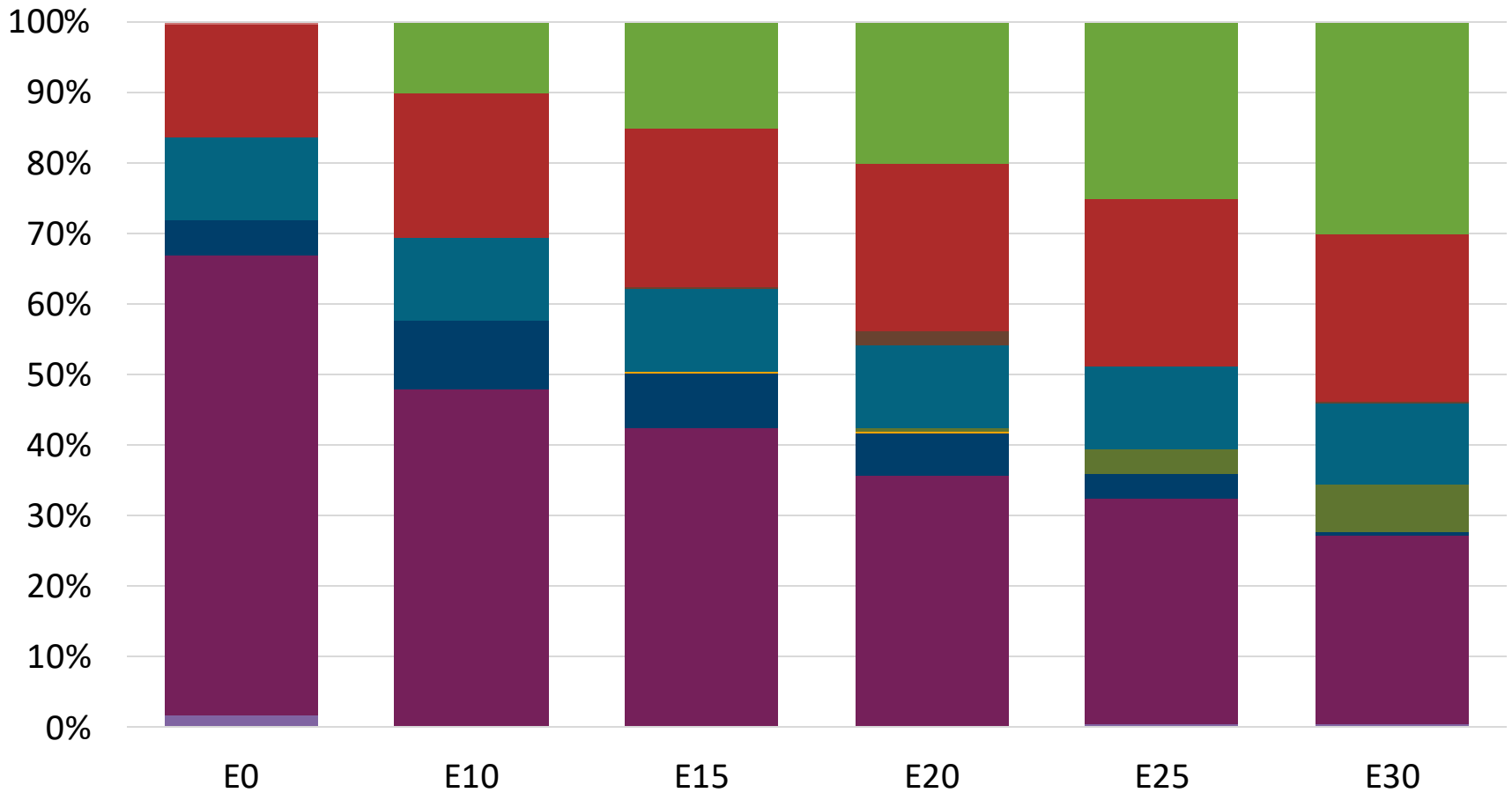
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components

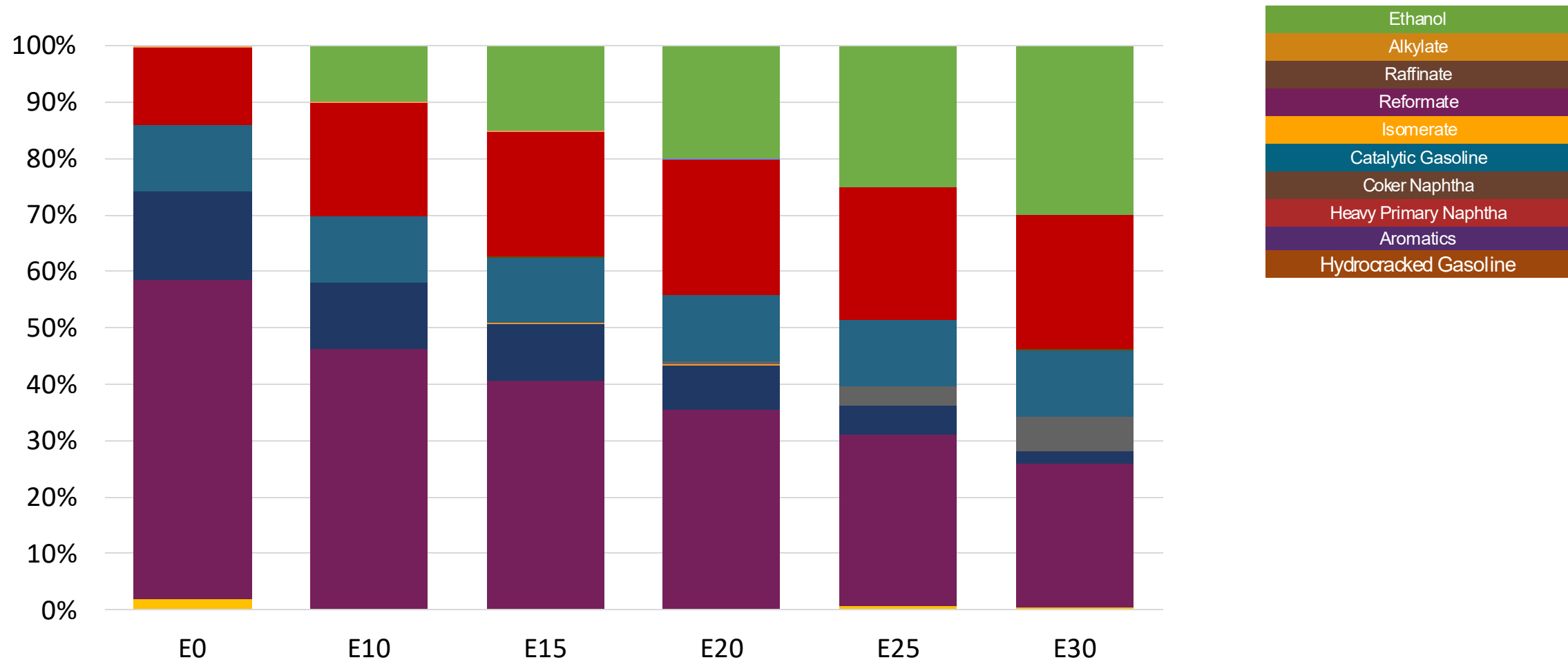
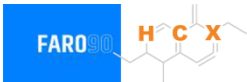


Argentina – Nafta Grado 2 – North & Central Zones– Constant Octane



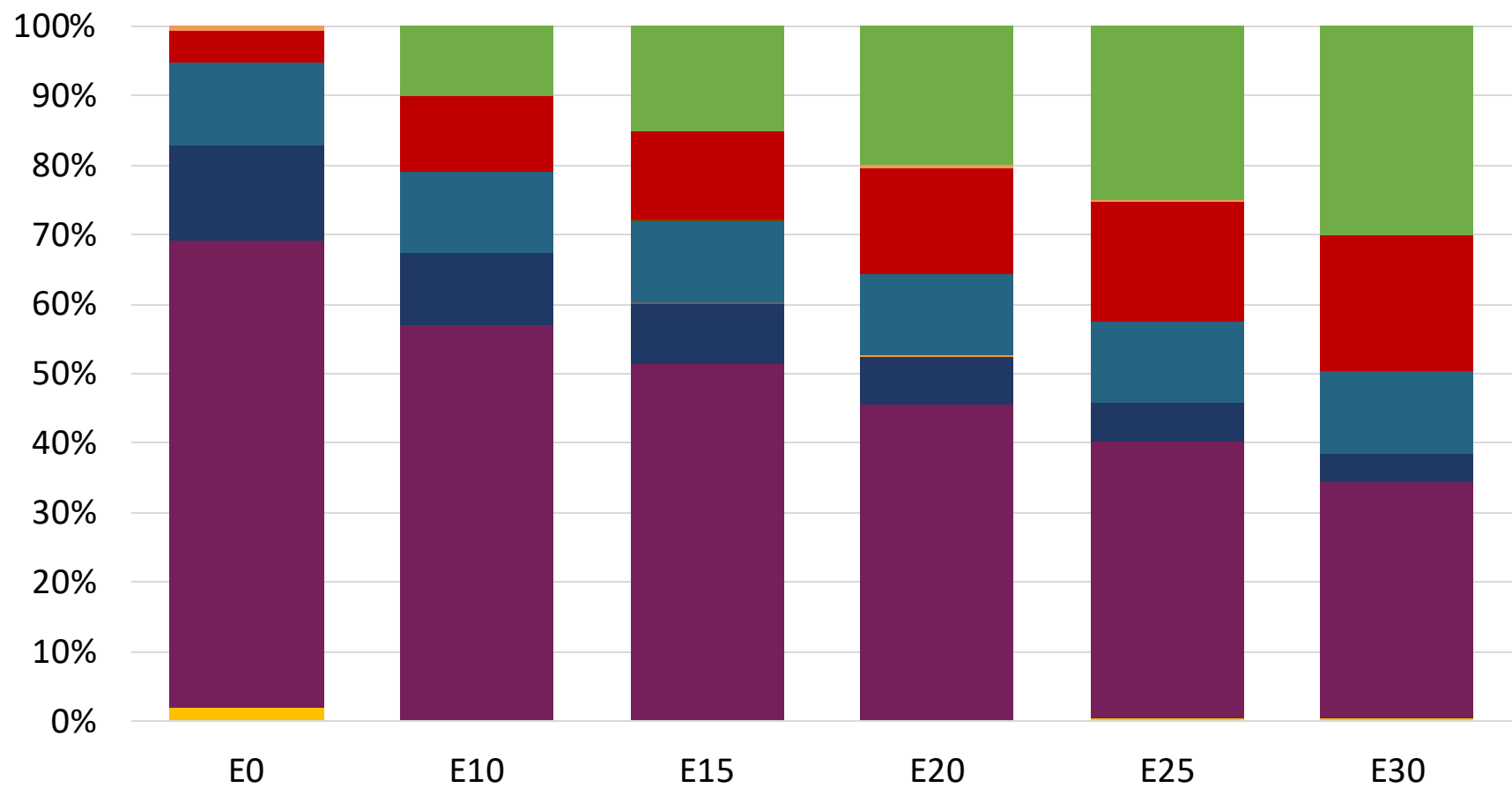
Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US\$/gal)	\$ 2.28	\$ 2.04	\$ 1.98	\$ 1.92	\$ 1.86	1.79

Argentina – Nafta Grado 2 – South Zone – Constant Octane



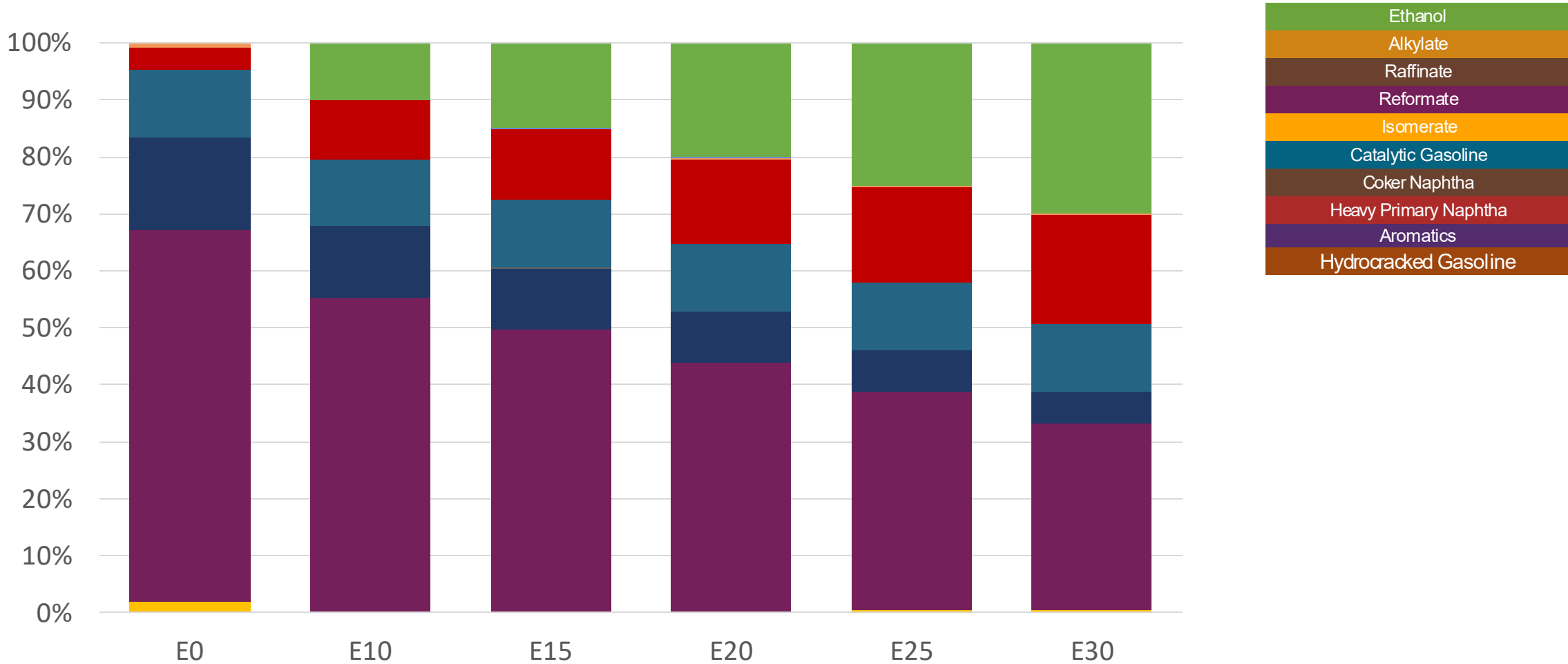
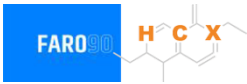
Octane (RON)	E0		E10		E15		E20		E25		E30	
	93.0		93.0		93.0		93.0		93.0		93.0	
Price (US\$/gal)	\$	2.13	\$	2.01	\$	1.95	\$	1.89	\$	1.83		1.77

Argentina – Nafta Grado 3 – North & Central Zones– Constant Octane



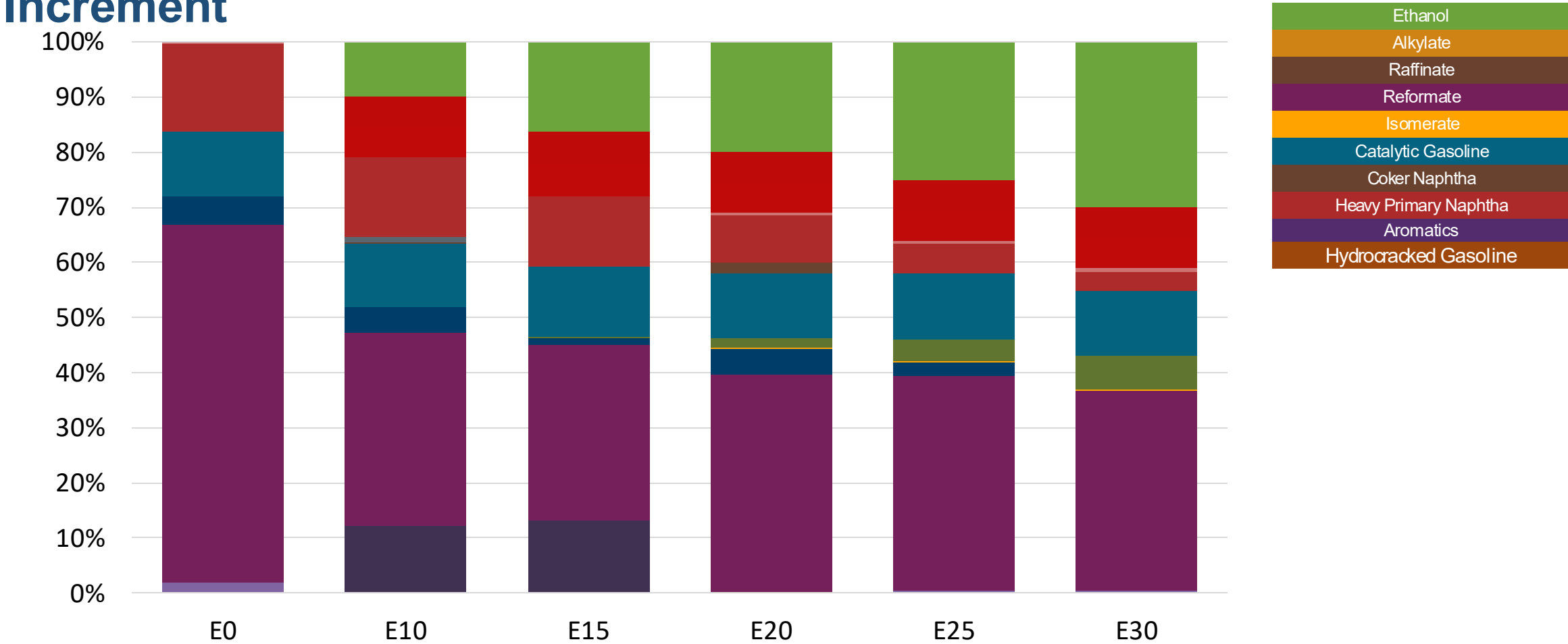
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.35	\$ 2.22	\$ 2.16	\$ 2.10	2.03	\$ 1.97

Argentina – Nafta Grado 3 – South Zone – Constant Octane



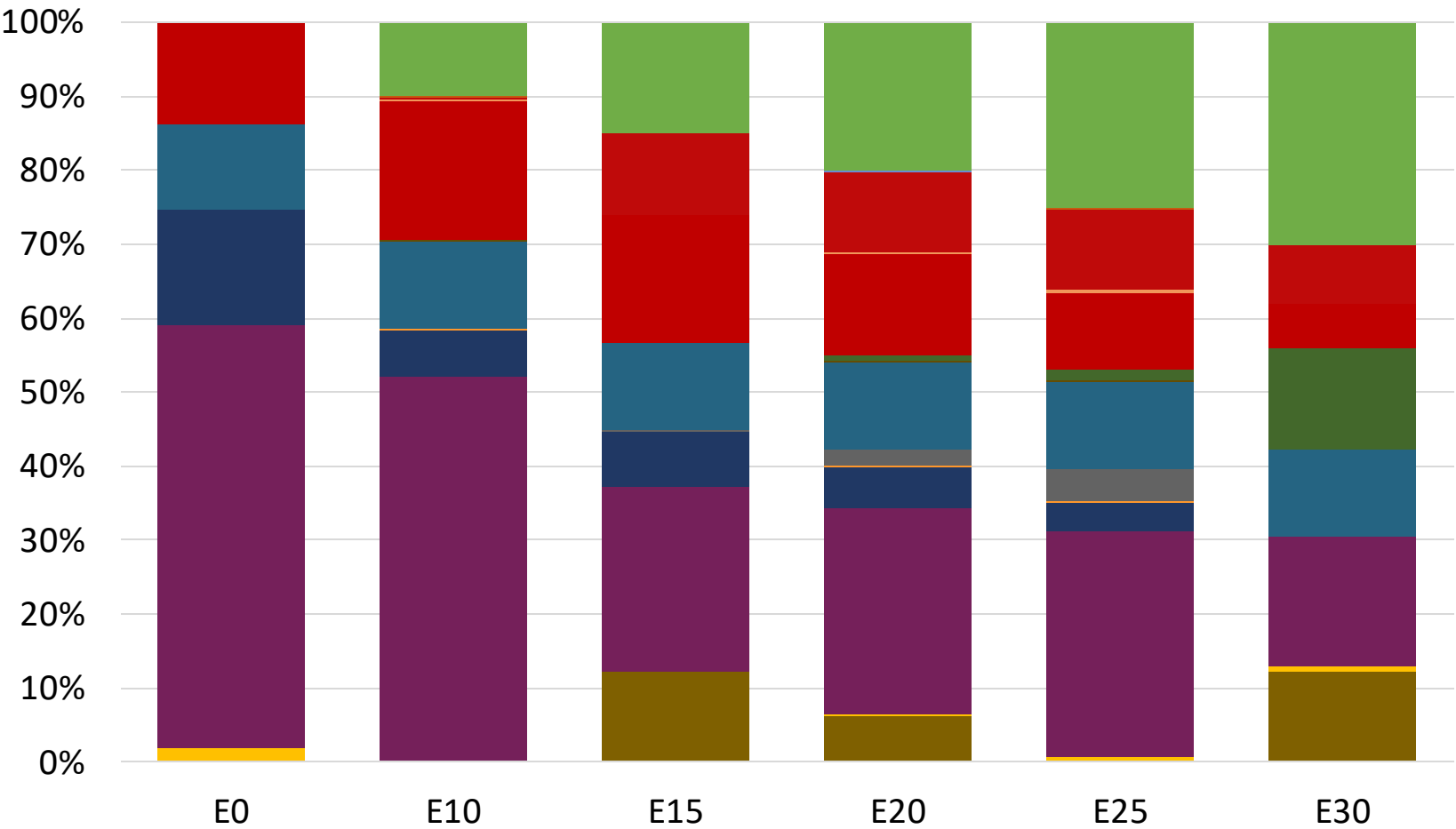
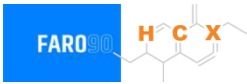
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.32	\$ 2.19	\$ 2.13	\$ 2.07	2.01	\$ 1.95

Argentina – Nafta Grado 2 – North & Central Zones– Octane Increment



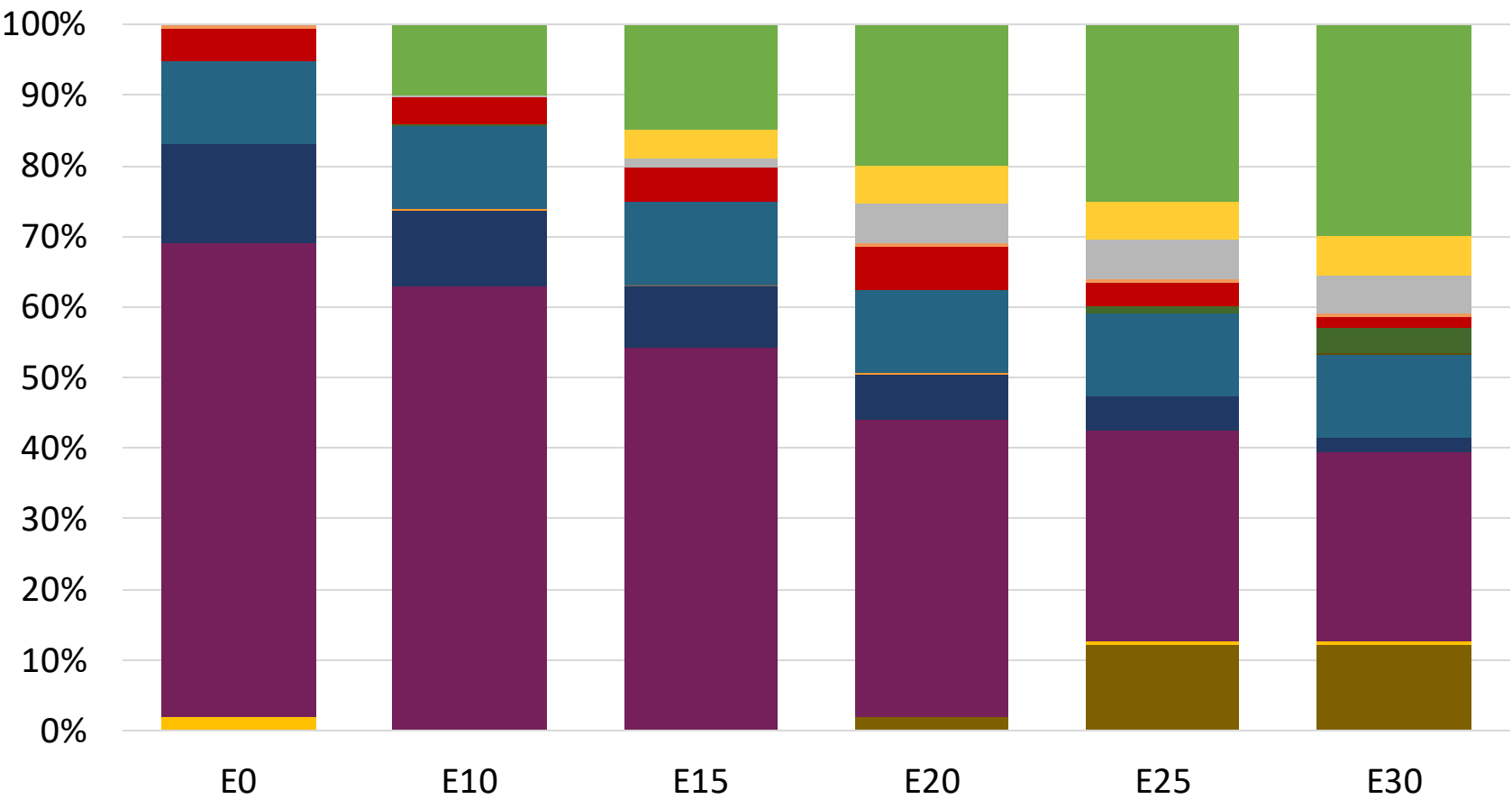
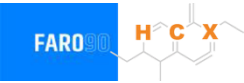
Octane (RON)	93.0	95.4	96.0	99.9	101.3	102.6
Price (US\$/gal)	\$ 2.28	\$ 2.28	\$ 2.28	\$ 2.28	\$ 2.28	\$ 2.28

Argentina – Nafta Grado 2 – South Zone – Octane Increment



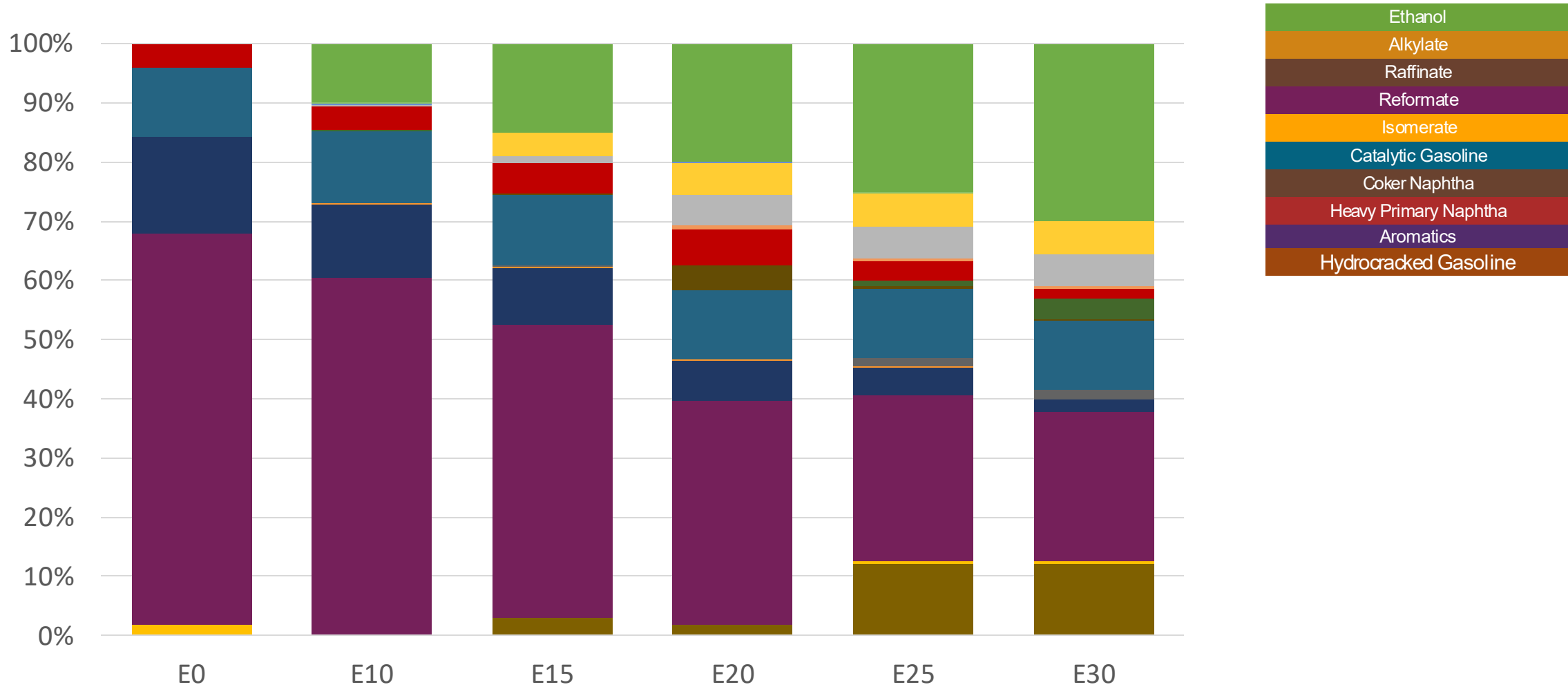
Octane (RON)	93.0	93.9	94.9	96.8	98.6	98.1
Price (US\$/gal)	\$ 2.13	\$ 2.13	\$ 2.13	\$ 2.13	\$ 2.13	\$ 2.13

Argentina – Nafta Grado 3 – North & Central Zones– Octane Increment



Octane (RON)	97.0	99.8	100.8	101.6	102.2	103.2
Price (US\$/gal)	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35

Argentina – Nafta Grado 3 – South Zone – Octane Increment



Octane (RON)	97.0	99.5	100.2	100.9	101.6	102.6
Price (US\$/gal)	\$ 2.32	\$ 2.32	\$ 2.32	\$ 2.32	\$ 2.32	\$ 2.32

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

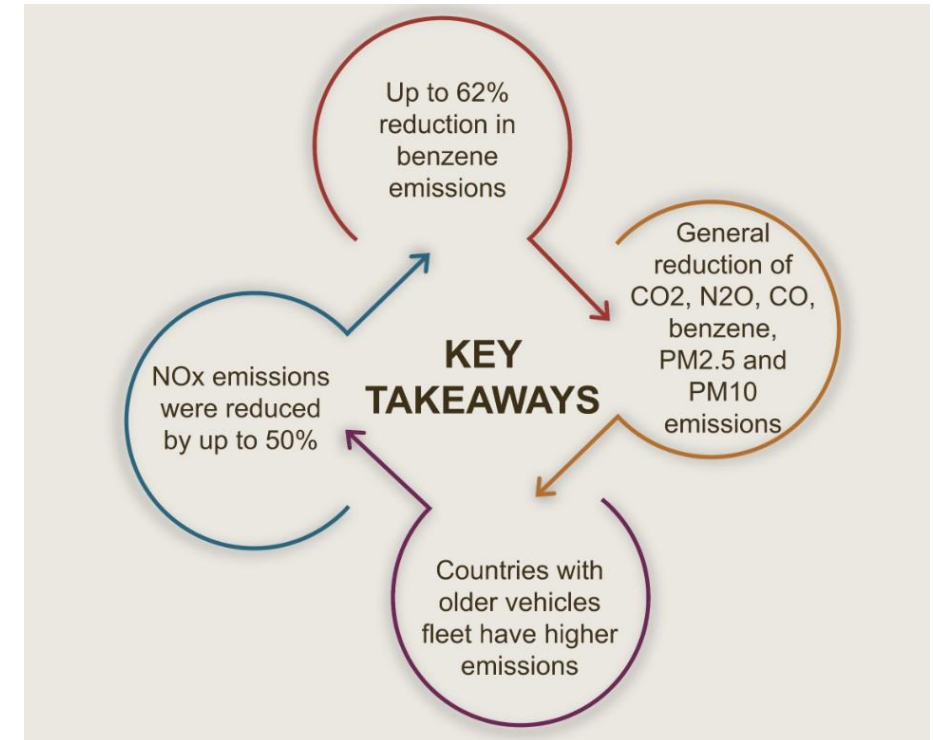
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

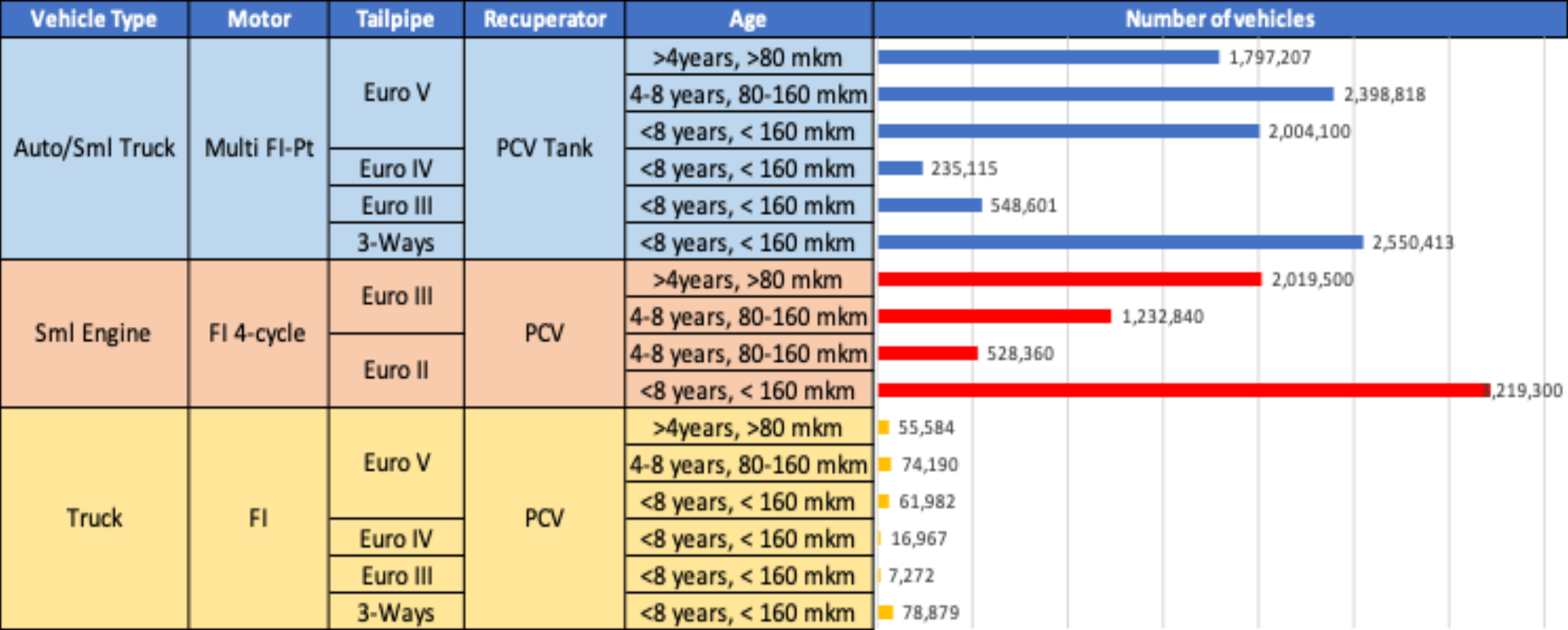
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



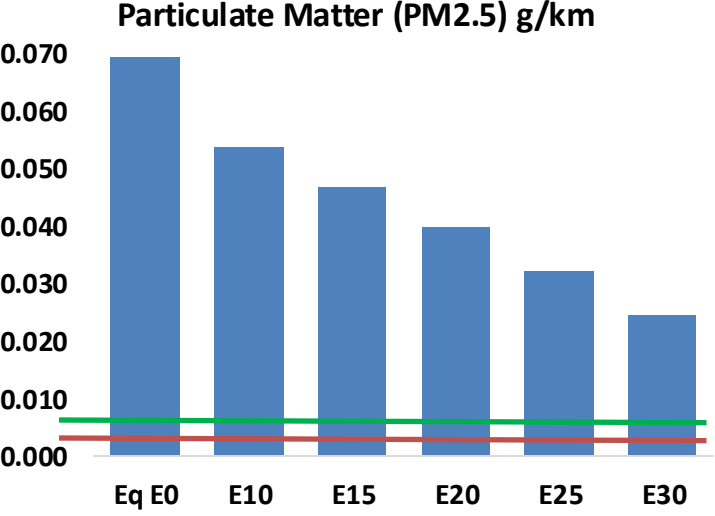
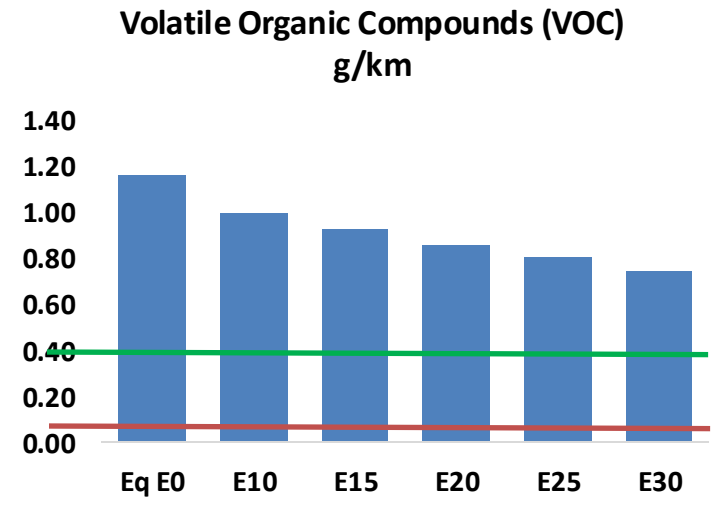
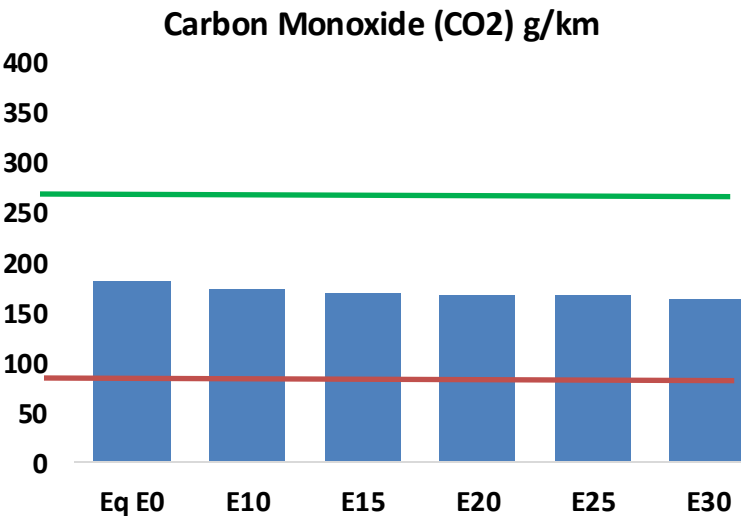
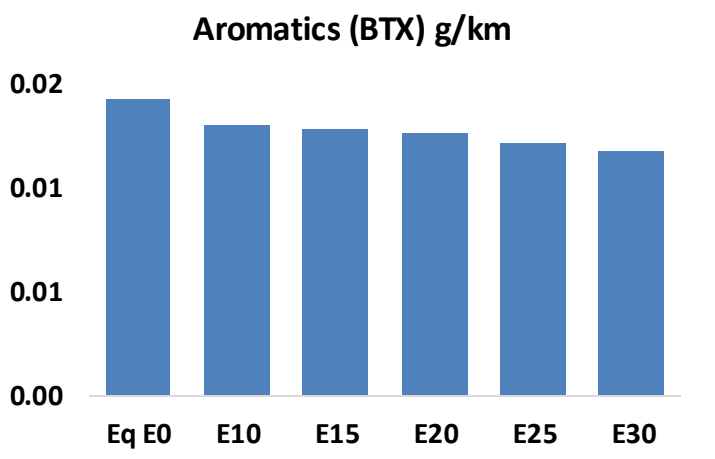
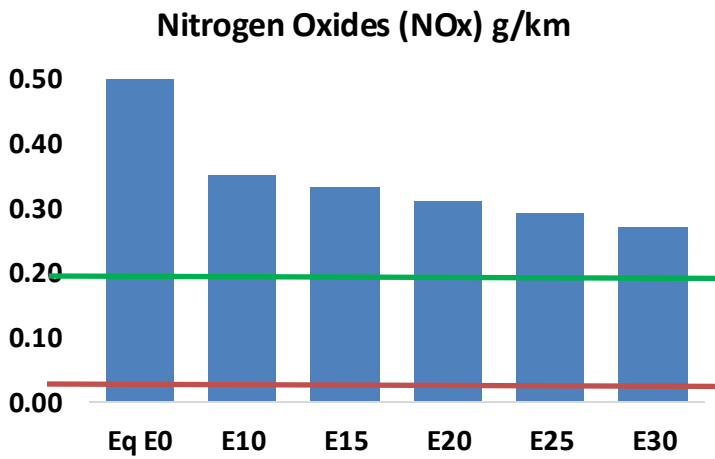
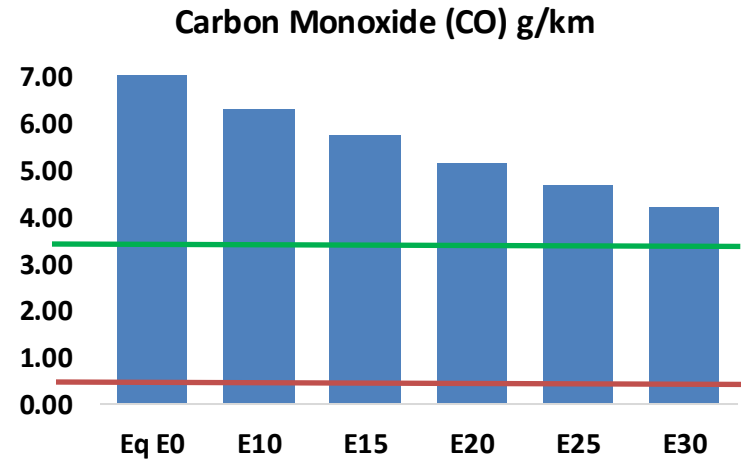
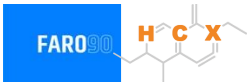
Gasoline Vehicle Fleet - Argentina



Vehicular Fleet: **16,829,128**
Average age: **12,5 years**
Motorcycles: **41.6%**

Argentina – Vehicular Emissions

United States
Euro 6 Standard



Argentina – Vehicular Emissions

Vehicular Emission (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,491,141	2,270,416	2,049,692	1,863,118	1,672,739	1,525,556	1,369,059	-18%	-33%
CO2	59,094,716	57,575,137	56,139,980	55,011,271	54,452,741	54,228,972	53,229,376	-5%	-8%
VOC	376,283	349,367	322,451	300,787	279,167	262,597	240,292	-14%	-26%
NOx	163,026	122,269	114,118	107,556	101,435	94,685	87,545	-30%	-38%
SOx	701	648	595	551	508	461	412	-15%	-28%
NH3	25,841	25,585	25,329	25,450	25,736	25,937	26,105	-2%	0%
N2O	1,094	1,088	1,084	1,089	1,112	1,124	1,137	-1%	2%
CH4	82,636	82,636	82,636	83,875	85,693	87,263	88,751	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,740	2,575	2,411	2,285	2,163	2,070	1,942	-12%	-21%
Acetaldehydes	7,767	8,914	13,079	19,918	27,136	31,008	36,639	68%	249%
Formaldehydes	30,818	29,962	34,927	41,012	42,966	47,532	51,744	13%	39%
Benzene	4,625	4,445	4,212	4,145	4,111	3,931	3,804	-9%	-11%
PM2.5	4,146	3,681	3,217	2,793	2,372	1,925	1,459	-22%	-43%
PM10	18,312	16,260	14,208	12,338	10,477	8,502	6,443	-22%	-43%

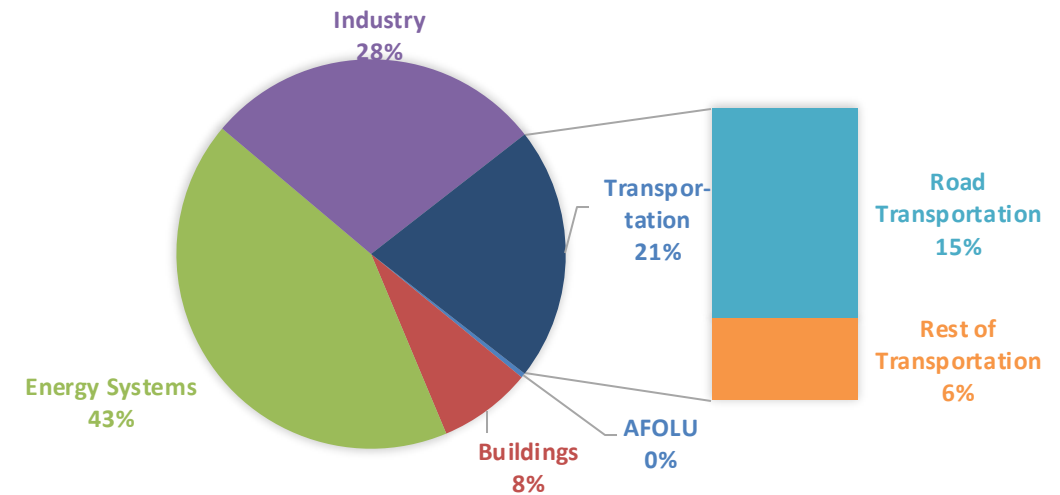
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

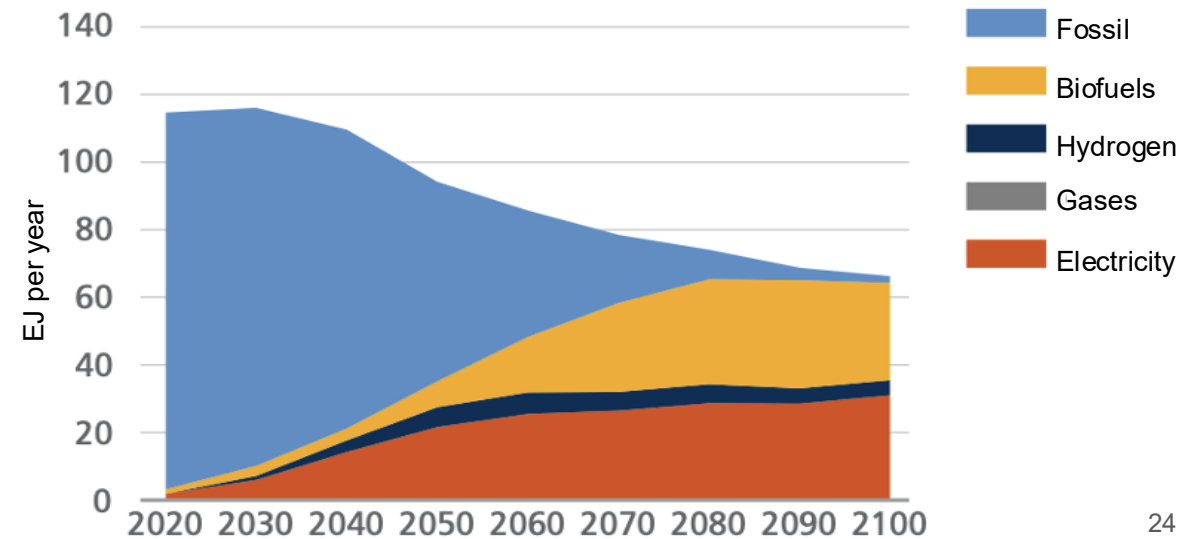
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution

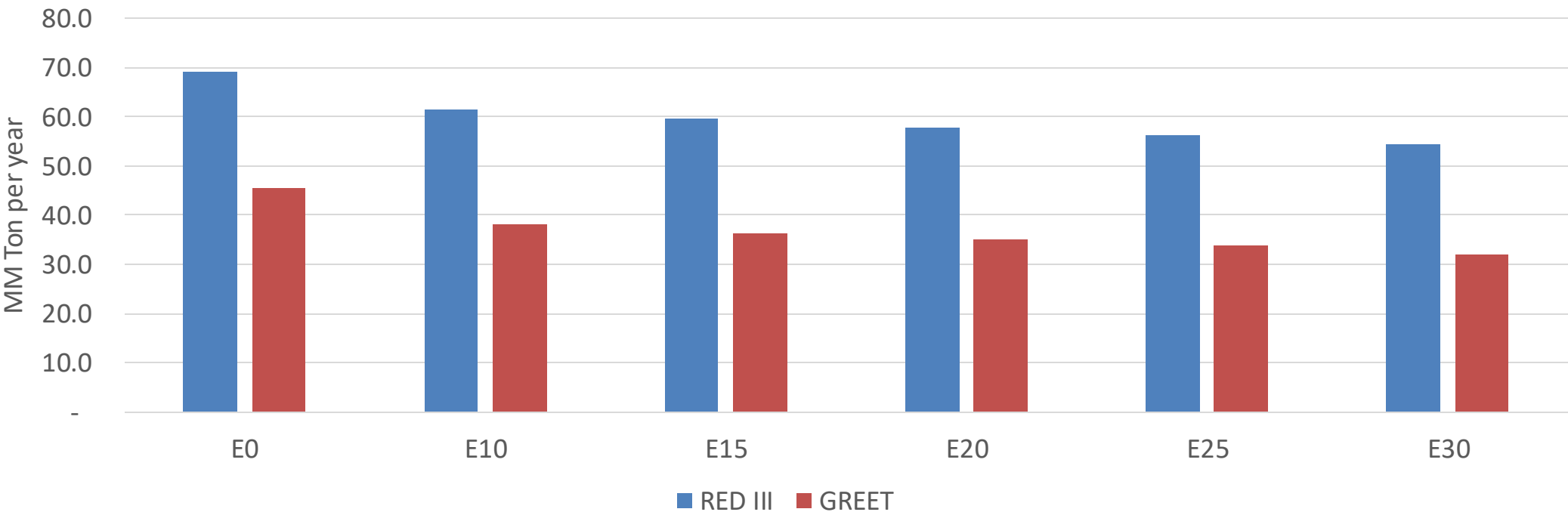


IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Argentina – Gasoline Vehicle Fleet GEI Emissions

Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2024 (MMT/año)	359.7
Target @ 2035 (MMT/year)	197.5
Delta	162.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0%	16%	20%	23%	25%	29%
RED II/III	0%	11%	14%	16%	19%	21%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.5%	5.5%	6.4%	7.2%	8.2%
RED II/III	0.0%	4.9%	6.0%	7.0%	8.0%	9.1%